True/False Questions (10 pts)
Problem 1.1: False (should be non linear function)
Problem 1.2: False (should computationally heavy but memory efficient)
Problem 1.3: True
Problem 1.4: False
Problem 1.5: True

Adalines (15 pts)

Problem 2.1:

x1	**x2**	**s**	**y**
-1	-1	-3	-1
-1	+1	-1	-1
+1	-1	-1	-1
+1	+1	+1	+1

Logic: AND

Problem 2.2:

x1	**x2**	**s**	**y**
-1	-1	+3	+1
-1	+1	+1	+1
+1	-1	+1	+1
+1	+1	-1	-1

w0 = +1, w1 = -1, w2 = -1

Problem 2.3

x1	**x2**	**x3**	**s**	**y**
-1	-1	-1	-3	-1
-1	-1	+1	-1	-1
-1	+1	-1	-1	-1

-1	+1	+1	+1	+1
+1	-1	-1	-1	-1
+1	-1	+1	+1	+1
+1	+1	-1	+1	+1
+1	+1	+1	+3	+1

w0 = 0, w1 = +1, w2 = +1, w3 = +1

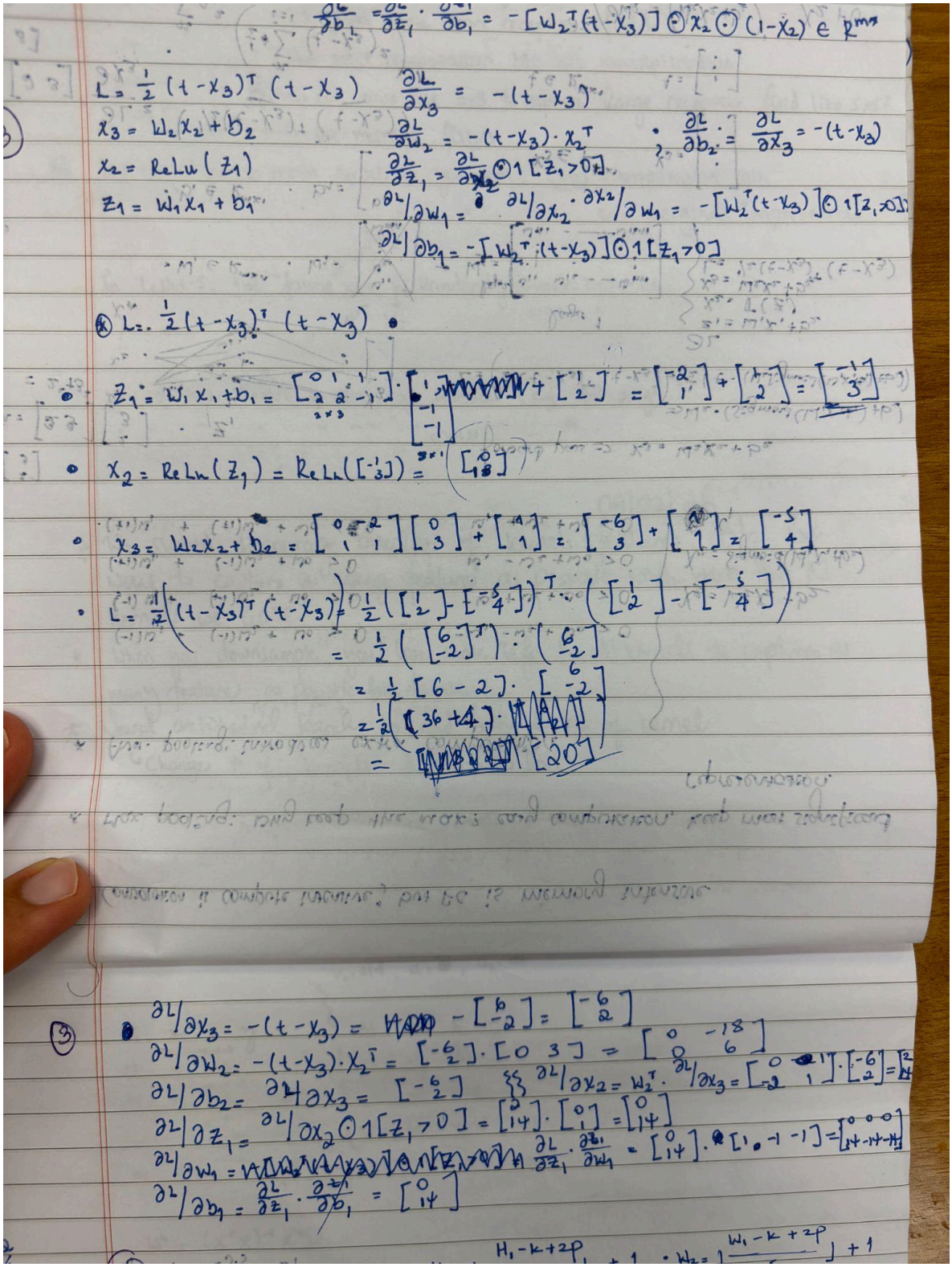
Problem 2.4

x1	**x2**	**s**	**y**
-1	-1	-1	-1
-1	+1	+1	+1
+1	-1	+1	+1
+1	+1	-1	-1

w20 = +1, w21 = +1, w22 = +1

Back Propagation (15 pts)

Problem 3.1 and Problem 3.2



2D Convolution
Problem 4.1

ece_661_hw1_4.ipynb

Problem 4.2

ece_661_hw1_4.ipynb

Lab: LMS Algorithm

SimpleNN.ipynb

Lab: Simple NN

SimpleNN.ipynb

w2 = [(w1 + 2p - k) / 2] + 1

If H * W = 32 * 32, P = 0 and S = 1
[(32 + 0 - 5) / 1] + 1: 28
[(32 + 0 - 10) / 1] + 1: 23

If H * W = 32 * 32, P = 2 and S = 1
[(32 + 4 - 5) / 1] + 1: 32
[(32 + 5 - 7) / 1] + 1: 31